The data model

Trunk section: §5 The data architecture.

Every table of the new desk in one place: its owner (the only writer), its key, its columns, and the legacy store it replaces. The domain leaves describe what each table means; this leaf is the schema and the map from the old data.

Built on: the three legacy data models, screeners, qp and journal, and the contracts section of every design leaf.

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1. Conventions

Rule	Meaning
one writer	every table names its owner; no other component writes it
instants	`timestamptz`, stored in UTC
trading days	`date`, the New York date
clock times	`HH:MM` New York, only for configuration (capture minutes, windows)
JSON	`jsonb` for documents that are read whole (a card, a strategy, rules)
honest null	a missing value is `NULL` in a column, or a `status` with a `reason` on a fact; never 0
versions	definitions that change (screeners, strategies, settings, models, the photo) carry a version; results record the version that produced them
migrations	numbered, run once, recorded; never a start-up repair
symbols	canonical ticker (no exchange prefix); `tv_symbol` kept beside it where TradingView needs it

2. The tables, by domain

Platform (PLATFORM.md)

Table	Key	Columns
`trading_calendar`	day	`is_open`, `open_at`, `close_at` (half days), `note`
`users`	id	name, password hash (scrypt), created
`sessions`	id	user, created, expires, last seen, user agent
`jobs`	job	domain, schedule, calendar, depends_on, enabled, expected_within, max_duration
`job_runs`	job, started_at	ended_at, result (`ok`, `failed`, `missed`, `skipped`, `hung`), counts (jsonb), error, stack
`settings`	key, scope	value, type, range, default, description, version
`setting_versions`	key, scope, version	value, changed_at, changed_by
`alerts_sent`	id	signal, sent_at, message, recovered_at
`backups`	night	manifest (jsonb), verified, row counts, size

Sources (SOURCES.md)

Table	Key	Columns
`source_calls`	adapter, day	calls, failures, last_error, last_error_at, p50 / p95 duration
`health_checks`	check, at	severity (`ok`, `degraded`, `down`), detail, fix

Bars live in the Parquet cache (per feed, ticker, timeframe), outside the database, with its limits.

Tools and screeners (TOOLS-AND-SCREENERS.md)

Table	Key	Columns
`tools`	id	name, description, accent, status (running, paused, archived, deleted), paused_at, paused_reason, capture_r1, entry_a, entry_b, capture_why, role (method, control, benchmark), created_at
`screeners`	tool, key	name, status, mirror_of, purpose (cards, list), direction (long, short, either)
`screener_versions`	tool, key, version	rules (jsonb), sort, limit, run_window, check_window, floor, post_tests (jsonb), created_at, note
`scan_reports`	tool, at	screeners asked, rows, drops by reason (jsonb), candidates, new cards, ms, errors

Card (CARD.md)

Table	Key	Columns
`facts`	subject, family, known_at	value (jsonb), status (`ok`, `none`, `unavailable`), reason, source, as_of, provisional
`facts_latest`	subject, family	the same columns: the newest fact of that family, 5-minute quotes included
`cards`	ticker, day	the latest projection (jsonb) of every family, `market_state_id`, updated_at
`tool_cards`	tool, ticker, day	screener_matches (jsonb: key, version, first match), first_seen, live, last_matched_at, manual_bias, score (jsonb: question, value, model version, sample, interval, breakdown)

`facts` and `facts_latest` are the fact store of the trunk (§5.2), shared by the card, market and research domains; each family has one writer (§3), which writes both. Every fetch updates `facts_latest`; `facts` keeps every fetch except for the fast families (`quote`, `derived`, `market_context`), whose history holds only discovery, each photo and the close. A tool card’s star is its `shortlist` row, not a column.

Market (MARKET.md)

Table	Key	Columns
`market_states`	id	at, day, indexes, sectors, short_term, mid_term, long_term, regime, oneil_m, oneil_l_ref, status, definition version (each jsonb where structured)
`hot_sector_days`	sector, day	best_score, days_at_40, days_under_20, hot, hot_since

Registers (REGISTERS.md)

Table / view	Key	Columns
`r1_photos`	tool, day, ticker	captured_at, late (taken after the capture minute: kept out of training unless included), card (jsonb), tool_card (jsonb), market_state_id, capture_version
`r2_slots` (view)	tool, day, slot	from `market_states`
`bars_day`	ticker, day	1-minute and daily bars used for outcomes (or a pointer to the Parquet file)
`r3_outcomes`	tool, day, ticker, entry	entry_price, hh, ll, atr14, up_r, down_r, or_high, or_low, result, reason, captured_at
`r4` (view)	tool, day, ticker, entry	`r1_photos ⋈ r3_outcomes`
`r4_uploads`	tool, day, ticker, entry, source	the row (jsonb), uploaded_at
`shortlist`	tool, day, ticker	added_at, method, price, change, sector, score, exported_at

Learning (LEARNING.md)

Table	Key	Columns
`models`	tool, question, version	features, coefficients, scaler, training window, sample, walk-forward metrics, published, reason, trained_at
`scorecards`	tool, screener key, screener version, entry, computed_on	cards, days, good %, interval, big %, median R, average R, adverse R, days hit, verdict, thresholds version

Quant (QUANT.md)

Table	Key	Columns
`strategies`	id	name, current version, stage, tools
`strategy_versions`	id, version	document (jsonb), created_at
`backtests`	id	name, spec (jsonb), strategy versions, status, progress, summary, error, created_at, updated_at
`backtest_trades`	id	backtest, day, symbol, side, entry_at, exit_at, entry, exit, ret, reason, ctx (jsonb: the register columns of its card)
`alerts`	strategy, symbol, bar_at	strategy version, sent_at, pushed
`primitive_approvals`	primitive, commit	approved_by, approved_at, notes (mirrors `approvals.json`)

Journal (JOURNAL.md)

Table	Key	Columns
`accounts`	id	name, broker, paper, every_minutes, look_back_days, enabled, last_run (jsonb)
`fills`	account, external_id	at, symbol, side, qty, price, fees, order_id, source
`trades`	id	account, symbol, direction, entry_at, exit_at, entry, exit, shares, stop, target, gross, fees, net, status, setup, notes, tags, touched_by_trader
`trade_fills`	trade, fill	portion
`setups`	id	name, description, enabled
`trade_snapshots`	trade	facts as of the entry (jsonb), market_state_id, r1 reference, mae, mfe
`day_notes`	day	note, updated_at

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3. The fact families

Every family is a row type in `facts`. Its writer, its subject and its clock:

Family	Subject	Writer	Clock
`quote`	ticker	card engine	5 minutes (latest); history at discovery, each photo and the close
`derived`	ticker	card engine	with the quote (history as the quote's)
`short_interest`	ticker	card engine	daily
`borrow`	ticker	card engine	daily
`reg_sho`	ticker	card engine	settlement day
`news`	ticker	card engine	each scan, live cards
`catalyst`, `bias`	ticker	card engine	with news / quote
`themes`	ticker	card engine	daily
`move_tests`	ticker	card engine (discovery)	at discovery
`market_context`	ticker	card engine (from the market state)	each scan (history as the quote's; the market states keep their own)
`bars_universe`, `group_ranking`, `oneil_m`	`market`	research worker	each close; `oneil_m` also every 30 minutes in the session, marked provisional
`rs_rating`, `rs_line`, `sic`, `oneil_l`, `oneil_c`, `oneil_a`, `oneil_s`, `oneil_i`, `oneil_n_base`, `oneil_n_events`, `oneil_ratings`	ticker	research worker	per ONEIL.md §6.1

4. From the legacy stores

Every legacy store, and where its data goes. All of it is imported once, running and archived tools alike, and the import is checked by row counts (MIGRATION.md).

A screener tool's database (`data/tradedesk.db`, `data/t<n>.db`)

Legacy table	New home
`screeners`	`screeners` + version 1 of `screener_versions` (later edits unknown: recorded as such)
`settings` (live keys)	`settings` (scope: the tool); `tool.name`, `tool.paused` → `tools`
`settings` (31 dead keys)	not carried
`scheduler_jobs`	`jobs` (custom schedules kept)
`r0_checkpoint`	not carried (a live list; today's comes from tool cards)
`r1_frozen`	`r1_photos` (the card JSON split into card and tool card; no screener versions or market state id for old days: recorded as such)
`r2_market_snapshots`	`market_states` (old days, with the O'Neil status left blank where it was read at export: D-D3)
`r3a`, `r3b`	`r3_outcomes` (entry A, B)
`r4a_train`, `r4b_train` (source `capture`)	not carried: `r4` is a view
`r4a_train`, `r4b_train` (source `upload`)	`r4_uploads`
`shortlist`	`shortlist` (one row per item)
`analysis_model`	not carried (dead)

Shared files in `data/`

Legacy file	New home
`tools.config.json` + `tools.local.json`	`tools`
`tool-stages.json`	`tools`
`industry-map.json`	`facts` (`sic`), research worker
`oneil-market.json`, `oneil-groups.json`, `oneil-13f.json`, `oneil/`	`facts` (`oneil_*`), rebuilt by the research worker; the latest files imported as of their dates
`shortlist-all.json`	not carried (a query over `shortlist`)
`keys.json`, `r2.json`, `auth.json`, `journal-accounts.json`	the secret store (`auth.json` → `users`)

qp

Legacy store	New home
`~/.qp/platform.db` `strategies`	`strategies` + `strategy_versions` (version 1)
`backtests`, `backtest_trades`	`backtests`, `backtest_trades` (frozen strategy copies kept in `spec`)
`~/.qp/strategies/*.json`	nightly export (backup)
`chart/seeds/*.json`	versioned migrations
`approvals/approvals.json`	stays in git; mirrored to `primitive_approvals`
`.default-feed`	`settings`
`screener_sources.json`	not carried (registers are read from the database)
oplog	logs
`~/.qp-cache/` (bars)	the Parquet cache
`~/.qp-cache/grouped/`, `sic/`, `f13/`, `canslim/`	rebuilt by the research worker into `facts`

The journal (`data/journal/journal.db`)

Legacy table	New home
`journal_trades` (Alpaca)	`trades`, re-derived from their fills by the new pairing engine and compared (JOURNAL.md §6)
`journal_trades` (file imports, manual)	`trades` (with a note where the legacy pairing is known to be wrong: D-J1, D-J2)
`trading_setups`	`setups`
`journal_daily_notes`	`day_notes`
`journal_fee_profiles`	`accounts` (fee rule)
`journal_technical_snapshots`, `journal_context_snapshots`, `journal_conditions`, `journal_setup_templates`, `trade_cards`, `trade_card_checks`, `check_library`, `setup_check_assignments`	not carried (recomputed from the fact store where wanted)
the old app's tables (`r0`–`r4` copies, `trading_*`, `volume_baseline`, …)	not carried

Models

`src/scoring/outputs*` are not imported: the new models are trained from R4.

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5. Retention and size

Data	Kept	Note
`facts`	forever	the fast families keep three or so rows per ticker per day (discovery, photos, close); their 5-minute values live only in `facts_latest`
`facts_latest`	one row per subject and family	overwritten
`cards`, `tool_cards`	forever	about 1–2 MB a day for every tool together (estimate)
`market_states`	30 days at scan grain, then the R2 slots and the close per day	—
`r1_photos`, `r3_outcomes`, `shortlist`	forever	the research record
`job_runs`, `health_checks`, `source_calls`	90 days	—
`backtests`	until deleted	—
`trades`, `fills`	forever	—
bars cache	by size and free disk	rebuildable

The legacy backup was about 17 MB compressed a night for everything; the new database starts at roughly the same size and grows mainly with R1 photos (a few MB a month per tool) and the day’s cards (estimates).